Lung Histopathology Triage

AI-assisted triage for lung cancer screening in capacity-constrained health systems.

A working prototype. The classifier is 19 MB and runs entirely in a web browser on commodity hardware, with no GPU and no inference server.

Seed round $1M	**Scale round** $5M	**Non-dilutive** Cloud and HPC credits	**Model** 19 MB, in-browser

Pathology capacity is the constraint

- Screening programmes and ageing populations increase slide volumes. Pathologist numbers are not increasing at the same rate.
- Triage is manual. Every case waits for specialist review, including the large majority that prove benign.
- Time to diagnosis is governed by queue position rather than by disease biology, and it is the variable that most affects patient outcome.
- Laboratories in lower-resource settings carry comparable caseloads with a fraction of the specialist capacity, and are least served by software priced for well-resourced hospitals.

The opportunity is faster and cheaper routing of cases, so that specialist time is spent where it changes an outcome.

What the system does today

- End-to-end pipeline: slide upload, preprocessing, classification and an explainability overlay, presented in a single-page clinical interface.
- The model runs in the browser. It was exported to ONNX and verified client-side, so there is no GPU requirement, no inference server and no per-request inference cost.
- Numerical parity with the reference PyTorch implementation is verified at 3.8e-06 on identical input.
- The deployment artefact is a set of static files, so hosting and demonstration carry no meaningful marginal cost.

Validation status

Measured on 120 labelled LC25000 images, 40 per class.

66.7%	120 of 120	0 of 80	19 MB
3-class accuracy	benign vs malignant	adenocarcinoma vs squamous	model, in-browser

The model separates benign from malignant tissue reliably across all 120 samples. Adenocarcinoma and squamous cell carcinoma are not yet separated: all 80 malignant samples map to a single output, so the current model functions as a binary benign/malignant classifier.

These images come from the same public dataset used for training, so this is not a held-out result. Establishing a held-out split and publishing a validated accuracy figure is the first milestone of Phase 1.

Development status

Each item is scoped as work, with the Phase 1 deliverable named.

Area	Current status	Phase 1 deliverable
Subtype discrimination	Adenocarcinoma and squamous carcinoma are not separated by the current model (0 of 80 samples).	Retrained model with three-class discrimination measured on held-out data.
Validation	Figures are drawn from the training dataset. No held-out split exists yet.	Held-out and external validation, with a published confusion matrix.
Data	Public research dataset only.	Clinical data partnerships and a validated dataset of record.
Governance	Not yet initiated.	Data protection impact assessment, ethics submission and data-sharing framework.
Explainability	Occlusion sensitivity, used because browser inference has no gradient pass.	Benchmark against the reference method, plus a clinician usability study.
Regulatory	Not approved. The system is labelled not for clinical use.	Quality management system and a submission-ready technical file.

Every item above maps to a funded workstream

Area	Funded workstream	Verifiable outcome
Subtype discrimination	Architecture and loss redesign, with compute-intensive retraining.	Three-class accuracy on held-out data, published.
Validation	Curated splits, an external validation set and an evaluation harness.	Reproducible accuracy and confusion matrix, reviewable by third parties.
Data	Clinical data partnerships, labelling and quality assurance.	Signed agreements and a validated dataset of record.
Governance	Data protection impact assessment, ethics submission, data-sharing framework.	Approved governance pack, reusable across jurisdictions.
Explainability	Method comparison study against the reference approach.	Agreement metrics and a clinician usability study.
Regulatory	Quality management system and regulatory strategy.	Audited quality system and a submission-ready technical file.

Why the remaining work is capital-efficient

- The engineering risk is retired. The pipeline exists, it runs on commodity hardware, and its numerical correctness is verified.
- The model is 19 MB and inference is client-side, so there is no GPU fleet and no inference infrastructure to fund.
- Marginal cost per user approaches zero, which is what makes deployment viable in lower-resource health systems.
- The remaining cost is compute and validation rather than invention.
- Compute access is the highest-leverage form of support available, because donated credits and supercomputing allocation convert directly into validation throughput.

Use of funds

Staged so that a funder can stop after Phase 1 and still see a complete, verified result. Figures are planning targets, not quotations.

Phase	Ask	Focus	Milestone at exit
Phase 1	**$1M**	Subtyping model, held-out validation, governance pack, cloud and HPC compute.	Published three-class accuracy on held-out and clinical data.
Phase 2	**$5M**	Clinical pilots in capacity-constrained systems, regulatory submission, team and support.	Regulatory filing and validated deployment in a live pathway.
In kind	Credits and HPC	Training compute, validation sweeps, evaluation infrastructure.	A documented compute-to-result trail.

The ask

- Sovereign and development funds: $1M seed, followed by $5M at scale. The deliverable is a validated, deployable triage layer with measured impact on time to diagnosis in a screening pathway.
- Cloud and compute credit programmes: the fastest route to accelerating this work. Credits convert directly into validation throughput, are non-dilutive, and carry no execution risk for the provider.
- Supercomputing allocation: retraining for subtype discrimination is compute-bound. An allocation compresses a twelve-month model roadmap into a quarter.
- Impact measurement: quantifying patients reached and health system cost avoided in target deployment regions is a funded Phase 1 deliverable, and is reported against published methodology.

Phase 1 milestones

- Published three-class accuracy on held-out data, with the confusion matrix and evaluation script released alongside it.
- Subtype discrimination demonstrated at a clinically useful level.
- Validation on clinical data from partner centres, with governance approved and documented.
- Explainability benchmarked against the reference method and reviewed by pathologists.
- Quality management system in place and a regulatory submission strategy agreed.

The system remains labelled retrospective validation only, not for clinical use, until these milestones are met.